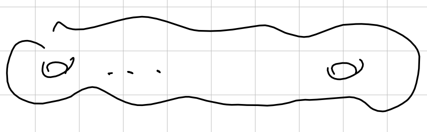
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



($n\geq 1$)

↓

$\pi_1=0$.

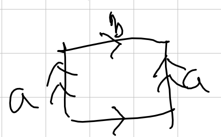
2-dim mfd

$L(p,q)$

$D^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

M∨N. ∧ ∨

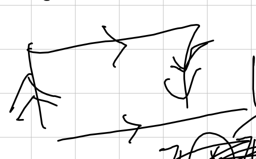
$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.



Klein. free product.

~~$\mathbb{Z}*\mathbb{Z}$~~

abababb..

$\langle a,b \mid aba=b\rangle$.

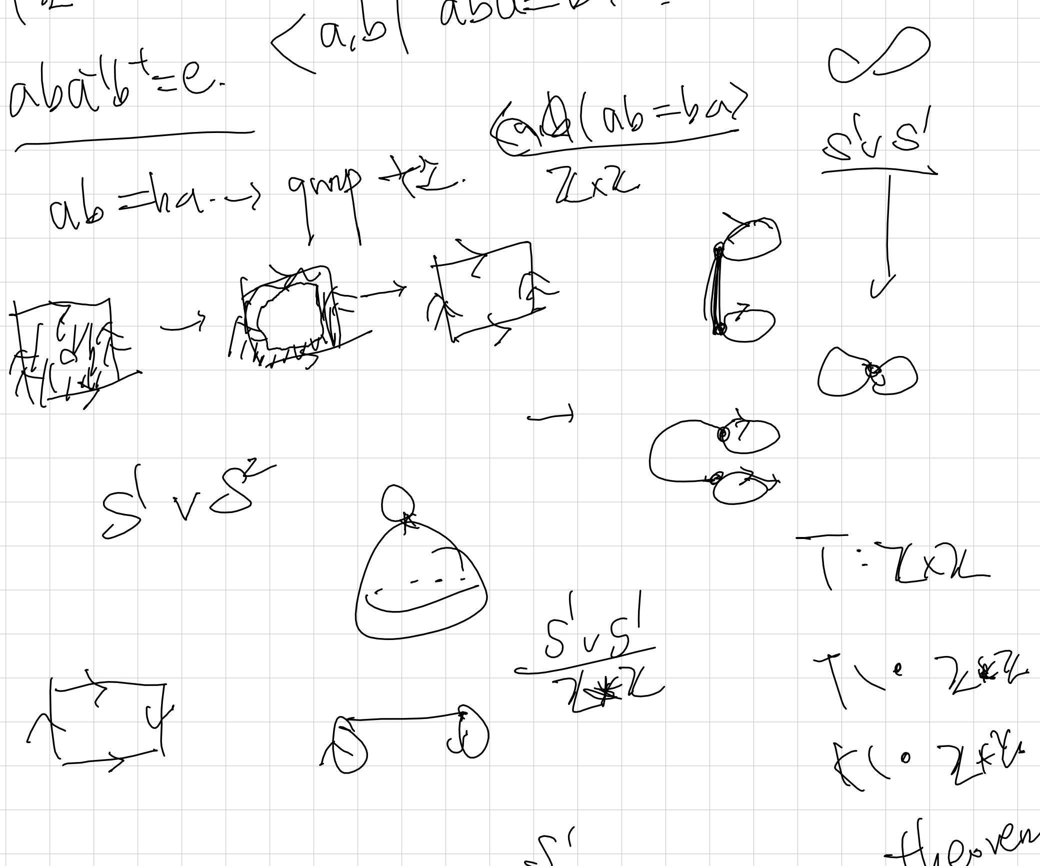
$aba^{-1}b^{-1}=e$.

$ab=ba$ → grup

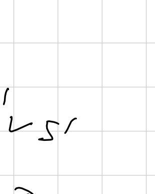
$\langle a,b \mid ab=ba\rangle$

$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$



$S^1\vee S^2$



$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$T: \mathbb{Z}\times\mathbb{Z}$

$T\setminus\cdot : \mathbb{Z}*\mathbb{Z}$

$K\setminus\cdot : \mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

S^1 ↕ $\mathbb{Z}$

S^3: 0.

↓

$\mathbb{R}^3$: 0.

M: n-dim

with $n\geq 3$

then $\pi_1(M)\simeq\pi_1(M\setminus\cdot)$.

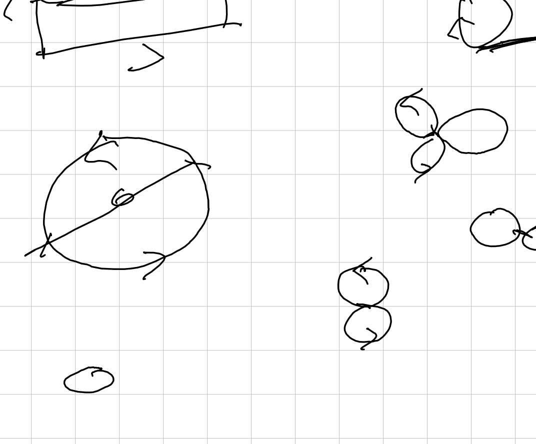
$\simeq\pi_1(M\setminus 2pt)$

$\simeq\pi_1(M\setminus\{x_i\})$.

$\pi_1(S^n)=0$ $n\geq 2$.

$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$ ($n\geq 2$).

tree



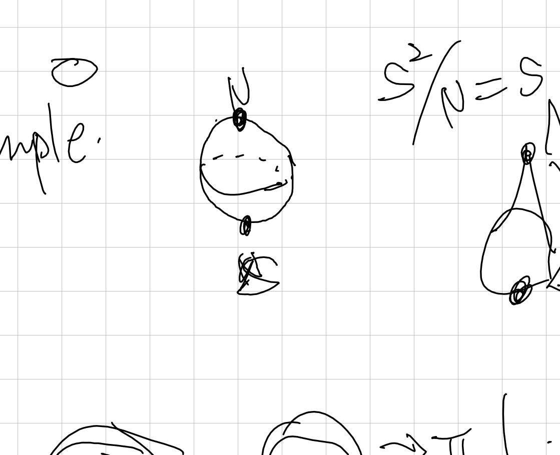
$S^1\vee S^1\vee\mathbb{R}$.

↓

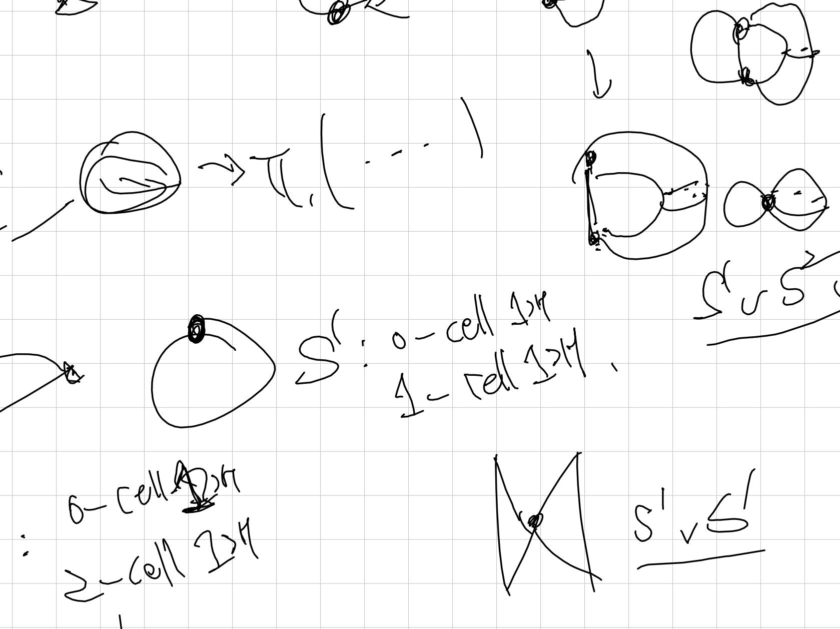
$\mathbb{Z}*\mathbb{Z}$

: $S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$



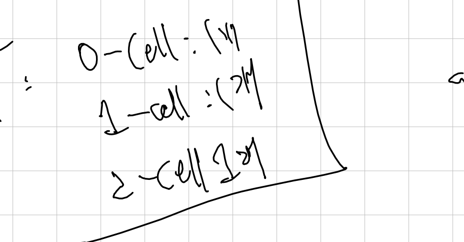
Example.



$S^2/N=S$

→ $\pi_1(\cdots)$

$S^1\vee S^2$,

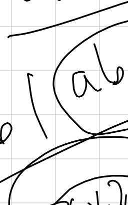


S^1: 0-cell 1개

1-cell 1개.

S^2: 0-cell 1개

2-cell 1개



$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell: 1개

1-cell: 1개

2-cell 1개

$aba^{-1}b^{-1}=0$.

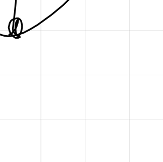
$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$

$\mathbb{Z}\times\mathbb{Z}$

$\mathbb{RP}^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$

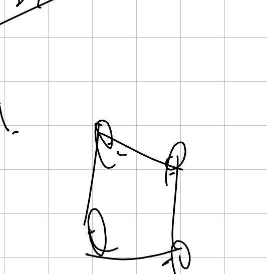
$a^2=1$



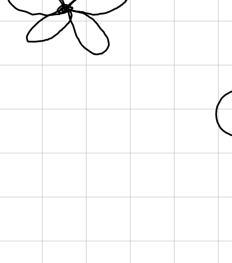
$\mathbb{RP}^3$: ~~0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

~~$\pi_1(\mathbb{RP}^4)$~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.



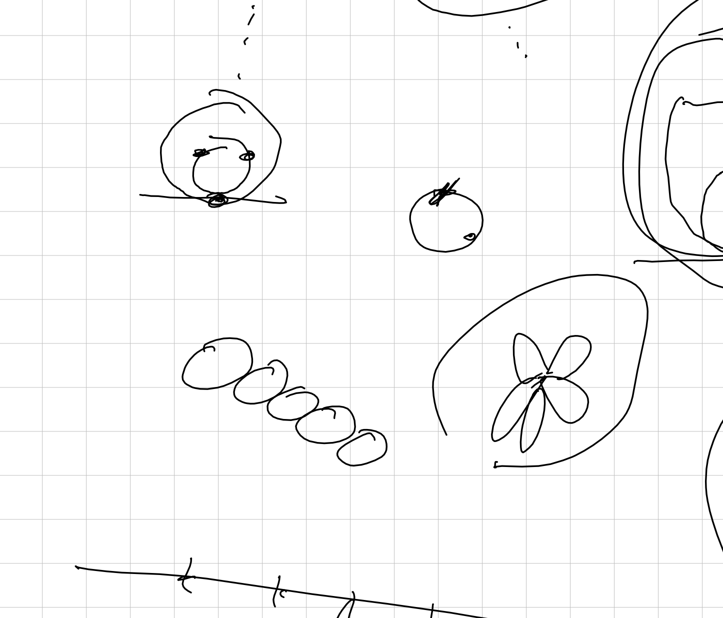
$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$



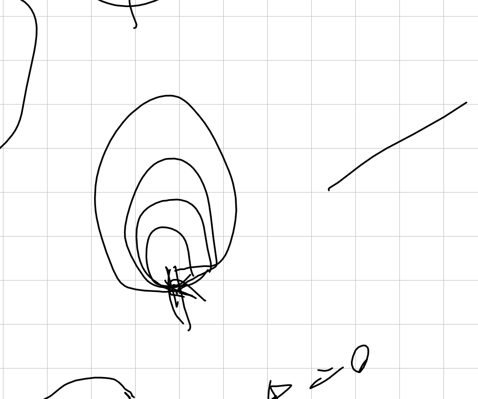
← $x^2+y^2+z^2=1$

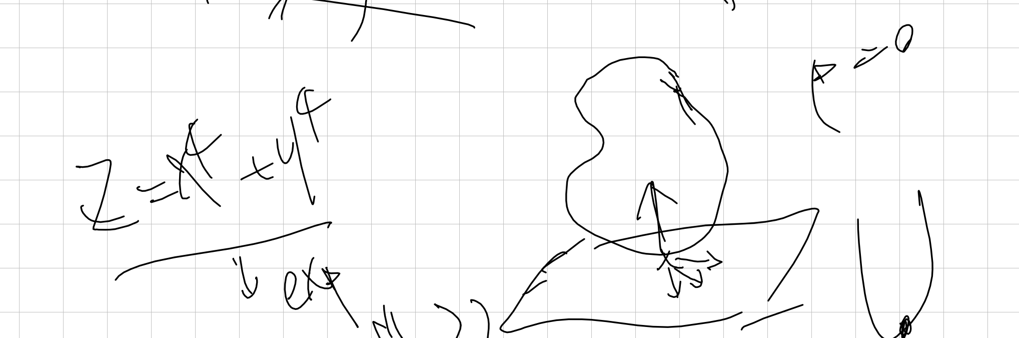
∪ $x^2+y^2\leq 1$

∪ $(0,0,z)$ $-1\leq z\leq 1$.

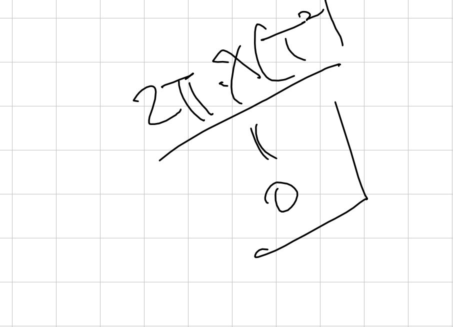


$\mathbb{R}/\mathbb{Z}$





$r=0$



$z=x^4+y^4$

↓ exp

$K_1=f(4,2)$..

$K_2=f(4)$

$\frac{1}{r^n+z}$.

$\int\frac{R\,dz}{y}$

$\int\frac{1}{z}dz$

$2\pi i$

$2\pi i\,\chi(\cdot)$

0